Do Now · retrieval

Client-side versus server-side processing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does an HTML form submit?

Q2

What does JavaScript textContent do?

Q3

What does GET generally do with form data?

Q4

A 2D array has 3 rows and 4 columns. Give total cells and last zero-based indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Client-side JavaScript can offer immediate feedback and responsive interfaces, but its validation can be bypassed or altered.

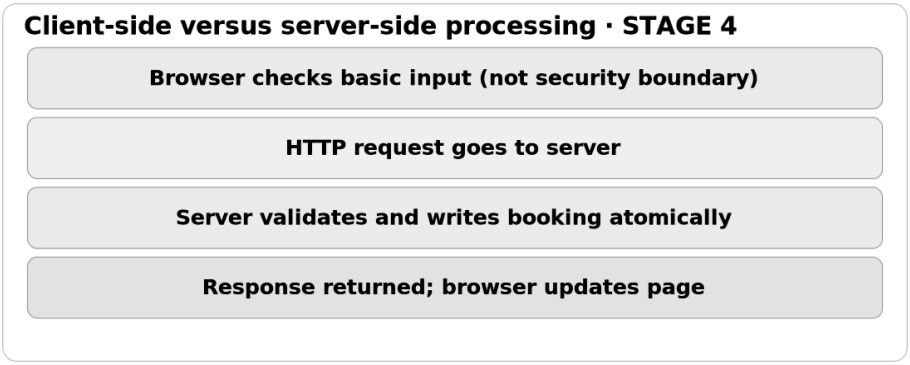
Core idea 2

Apply — Server-side code enforces permissions, validates input against business rules and updates authoritative records or databases.

Core idea 3

Evaluate — Processing choices trade off device load, server load, confidentiality, latency, accessibility and reliability; many systems use both.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

client-side — Processing on user's browser/device.

server-side — Processing on service infrastructure.

validation — Checking data against rules.

authentication — Verifying user identity.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

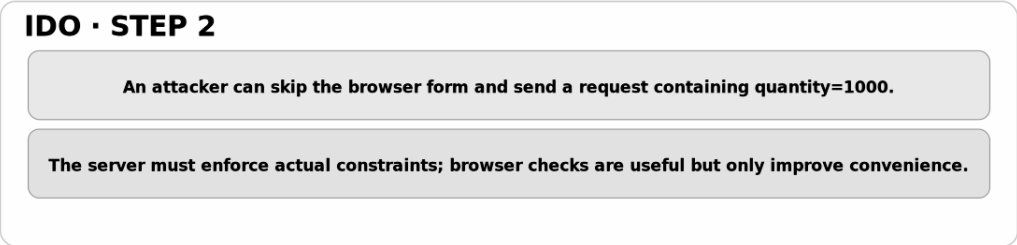
Why must a server check a submitted quantity even if the browser requires 1–5?

Follow the worked method

1. Predict first — An attacker can skip the browser form and send a request containing quantity=1000.

2. Explain — The server must enforce actual constraints; browser checks are useful but only improve convenience.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The browser can preview or resize an image to reduce traffic and improve usability.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Compare client-side and server-side processing using form validation. [6]

Question 2

An online shop must deduct stock and charge for an order. Explain which operations cannot rely solely on a browser. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate a hybrid design for an accessible online examination practice system. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A server can trust form data solely because the browser used required.

Use key vocabulary from memory

1. Explain client-side in your own words.

2. Explain server-side in your own words.

3. Explain validation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.